

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2025****Course: B. Tech – First Year****Branch : Common to All Branches****Semester : I/II****Subject Code & Name: BTES203 Engineering Mechanics****Max Marks: 60****Date:****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	In engineering mechanics, a body is said to be rigid if,	Remember	1
	a. it has large mass b. it does not deform under applied forces c. it is always stationary d. it has uniform density		
2	Newton's first law is also known as the law of	Remember	1
	a. force b. momentum c. inertia d. acceleration		
3	Friction is the force which	Remember	1
	a. assists motion b. opposes relative motion c. produces acceleration d. reduces weight		
4	The ratio of limiting friction to normal reaction is called	Remember	1
	a. friction angle b. coefficient of friction c. angle of repose d. sliding friction		
5	Newton's second law of motion states that force is equal to	Remember	1
	a. velocity × mass b. acceleration ÷ mass c. mass × acceleration d. momentum × time		
6	When a body is thrown vertically upward, its velocity at the highest point is	Remember	1
	a. maximum b. minimum c. zero d. equal to acceleration		

7	Which of the following equations represents D'Alembert's principle for translation?				Remember	1
	a. $\sum F = ma$	b. $\sum F - ma = 0$	c. $\sum M = I\alpha$	d. $\sum F = 0$ only		
8	The kinetic energy of a body of mass m moving with velocity v is				Remember	1
	a. mv	b. $\frac{1}{2}mv$	c. $\frac{1}{2}mv^2$	d. mv^2		
9	Impulse is defined as				Remember	1
	a. force $\times$ displacement	b. force $\times$ velocity	c. force $\times$ time	d. mass $\times$ acceleration		
10	The impulse–momentum principle states that impulse is equal to				Remember	1
	a. kinetic energy	b. change in momentum	c. work done	d. force applied		
11	A couple consists of				Remember	1
	a. two unequal parallel forces	b. two equal parallel forces acting in same direction	c. two equal parallel forces acting in opposite directions	d. two perpendicular forces		
12	If the resultant force acting on a body is zero, the body may be				Remember	1
	a. accelerating	b. in equilibrium	c. rotating with acceleration	d. unstable		
Q. 2	Solve the following.					12
A)	Classify the system of forces with neat sketches & explain them in detail?				Remember	6
B)	Find the magnitude and direction of the resultant R of four concurrent forces acting as shown in figure.				CO2	6

Q.3	Solve the following.		12
A)	A body of weight 1000 N is pulled upon inclined plane by a force of 600 N. The inclination of the plane is 27° to the horizontal and force is applied parallel to the plane. Determine the angle of friction.	CO2	6
B)	A semicircular area is removed from a trapezium as shown in figure (all dimension in mm). Determine the centroid of the remaining area (Shown hatched).	CO3	6
Q. 4	Solve Any Two of the following.		12
A)	Define: a) Static Friction, b) Dynamic Friction, c) Angle of Friction, d) Angle of repose. What are the Coulomb's laws of dry friction?	Remember	6
B)	A stone, dropped into well a well, is heard to strike the water after 4	CO3	6

	seconds. Find the depth of well. if velocity of the sound is 350 m/s.		
C)	Derive equation of the path of a projectile and write equation of time of flight, horizontal range, and maximum height of a projectile.	CO5	6
Q.5	Solve Any Two of the following.		12
A)	i) Define D'Alembert's Principle and write down its equation. ii) Define Newton's First, Second and Third Law of Motion.	CO1	6
B)	A particle, starting from rest, moves in a straight line, whose equation of motion is given by: $s = t^3 - 2t^2 + 3$. Find the velocity and acceleration of the particle after 5 seconds.	CO4	6
C)	If a particle is projected inside a horizontal tunnel which is 5 meters high with velocity of 60 m/s, find the angle of projection and the greatest possible range.	CO4	6
Q. 6	Solve Any Two of the following.		12
A)	State and prove work energy principle.	CO5	6
B)	An automobile is moving at a speed of 70 kmph, when the breaks are fully applied causing all four wheels to skid. Determine the time required to stop the automobile, a) On concrete road, coeff. of friction = 0.75, b) On ice for which coeff. of friction = 0.08	CO5	6
C)	A block weighing 2500N rests on a level horizontal plane for which coefficient of friction is 0.20. This block is pulled by a force of 1000 N acting at an angle of 30° to the horizontal. Find the velocity of the block after it moves 30m starting from rest. If the force of 1000 N is then removed, how much further will it move? Use work energy method.	CO5	6
	*** End ***		